



Vivekanand Education Society's Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering
Academic Year: 2025-26

Lab Code: NCMPC62	Year/ Semester : III/ Sem VI
Name of the Lab: Cryptocurrency & Blockchain Development	Class : D12A / D12B / D12C
Lab Teacher : Mrs. Geocey Shejy (D12A) Mrs. Lifna C. S. (D12B) Mrs. Pradnya Raut (D12C)	Subject Teacher: Mrs. Geocey Shejy (D12A) Mrs. Lifna C. S. (D12B) Mrs. Pradnya Raut (D12C)

Course Outcomes

1. Understand the principles, architecture, and applications of blockchain and cryptocurrencies.
2. Analyze and evaluate the features of major blockchain platforms and their use cases.
3. Design and implement secure cryptocurrency wallets and manage transactions effectively.
4. Develop and deploy smart contracts using Solidity and integrate them with blockchain networks.
5. Build and optimize decentralized applications (DApps) with cryptocurrency payment integration.
6. Assess and implement advanced blockchain solutions like DeFi, NFTs, and layer-2 scalability techniques.

Program Outcomes (POs)	
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).



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PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) Independent and life-long learning ii) Adaptability to new and emerging technologies and iii) Critical thinking in the broadest context of technological change. (WK8)

CO-PO Mapping :

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2
CO1	3	2	1	1	1	1	–	–	1	–	1	2	1
CO2	3	3	2	2	2	1	–	–	1	–	1	3	1
CO3	2	2	3	1	2	1	1	1	1	1	1	3	2
CO4	2	2	3	2	3	1	1	1	1	1	1	3	2
CO5	2	2	3	2	3	1	1	2	1	1	1	3	2
CO6	1	2	2	2	3	2	2	1	1	1	2	3	2



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Evaluation

- Experiments are evaluated based on mock viva taken on experiments.
- Journals are evaluated based on soft skills and presentations.
- Evaluation is based on the following table:-

Range	Grade
80 and above	Outstanding (O)
75.00 – 79.99	Excellent (A)
70.00 – 74.99	Very Good (B)
60.00 – 69.99	Good (C)
50.00 – 59.99	Fair (D)
45.00 – 49.99	Average (E)
40.00 – 44.99	Pass (P)
Less than 40.00	Fail (F)

No	Evaluation Component for Experiment	Marks
1	Program Execution	2
2	Documentation	1
3	Timely Checked	1
4	Viva	1
	Total	5 Marks
No	Evaluation Component for Case Study	Marks
1	Presentation	3
2	Documentation	2
3	Implementation / Execution with Viva	3
4	Completeness	2
	Total	10 Marks



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List of Experiments:

No	Title of Experiment	COs Mapped	Grade
1	Explore the Cryptocurrency Landscape	CO1	/ 5
2	Cryptography in Blockchain, Merkle root tree hash	CO2	/ 5
3	Create a Blockchain using Python	CO1, CO2	/ 5
4	Create a Crypto Currency using Python	CO1, CO2, CO3	/ 5
5	Hands on Solidity Programming Assignments	CO4	/ 5
6	Creating a Token (ERC-20) in Remix IDE	CO4, CO5	/ 5
7	Build a web based DApp using Truffle suite	CO4, CO5	/ 5
8	Configuring a Cryptocurrency Wallet using Metamask	CO3, CO5	/ 5
9	Simulating Blockchain Transactions using Ganache	CO2, CO3, CO4	/ 5
10	Implement a Private Ethereum Blockchain using Geth	CO2, CO4	/ 5

Continuous Assessment

1	CA -1 : Case Study (Presentation and Demonstration of Dapp)	CO1, CO2, CO3, CO4, CO5, CO6	/ 10
2	CA2 - MCQ1 (13th March 2026)	CO1, CO2, CO3	/ 5
3	CA2 - MCQ2 (20th April 2026)	CO4, CO5	/ 5
4	Other Rubrics mentioned in Syllabus (enlist them) - -		

Assignments

1	Assignment - 1 (DOP : 26th Feb 2026)	CO1, CO2, CO3	/ 5
2	Assignment - 2 (DOP : 20th April 2026)	CO4	/ 5

Faculty name:

Signature: